

1.1 Purpose

Document Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the NewPOPSys v1.38 platform, a PSP-led Point-of-Purchase (POP) Campaign Orchestration System. It serves as the definitive reference for all functional and non-functional requirements, system behaviors, and acceptance criteria for the v1 implementation.

Intended Audience

Audience	Usage
Development Teams	Primary implementation reference for building features, APIs, and user interfaces
AI Development Agents	Authoritative specification for agentic development workflows and autonomous code generation
Quality Assurance	Basis for test case development, acceptance criteria validation, and verification procedures
Product Management	Scope definition, feature prioritization, and stakeholder communication
System Architects	Technical design decisions, integration patterns, and infrastructure planning
Operations Teams	Deployment requirements, operational procedures, and support documentation
Project Stakeholders	Progress tracking, scope validation, and acceptance sign-off

Document Objectives

- 1. Define Complete Scope:** Establish clear boundaries for v1 functionality, explicitly documenting what is included and what is deferred to future releases.
- 2. Enable Agentic Development:** Provide structured, unambiguous specifications that support AI-assisted development with minimal human intervention.
- 3. Establish Traceability:** Create direct links between requirements, source SOW documents, wireframes, and acceptance criteria.
- 4. Support Verification:** Define measurable acceptance criteria and test scenarios for all functional requirements.
- 5. Prevent Scope Creep:** Document protected "Not-Now" features and guardrails to maintain project discipline.

Relationship to SOW Documentation

This SRS is derived from and complements the Statement of Work (SOW) documentation set:

Document	Version	Relationship
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Master SOW Compiled	v1.38	Primary source of scope and requirements
SUPP-001 through SUPP-039	Various	Detailed supplemental specifications
Screen Specifications	M/B/P/S Series	UI/UX requirements and wireframes
Database Model	DDL Files	Physical schema and data constraints
OpenAPI Specification	v1.0	API contract definitions

Document Conventions

Requirement Identifiers

All requirements use the following identifier format:

- **FR-XXX-NNN**: Functional Requirements (e.g., FR-CAM-001 for Campaign module)
- **NFR-XXX-NNN**: Non-Functional Requirements (e.g., NFR-SEC-001 for Security)
- **BR-XXX-NNN**: Business Rules (e.g., BR-VER-001 for Verification rules)
- **AC-XXX-NNN**: Acceptance Criteria (e.g., AC-CAM-001)

Module Prefixes

Prefix	Module
CAM	Campaign Management
STO	Store Management
ORD	Order/Fulfillment
EXE	Store Execution
VER	Verification
ISS	Issues/Reorders
EXP	Exports/Reports
INT	Integrations
SEC	Security/RBAC
PLT	Platform Operations

Priority Levels

Priority	Definition
P1 - Critical	Core vertical slice functionality; must be complete for pilot
P2 - Important	Required for full pilot operation; can follow critical path
P3 - Desired	Enhances user experience; may be deferred if schedule requires
P4 - Future	Documented for future reference; explicitly out of v1 scope

Status Indicators

Indicator	Meaning
LOCKED	Requirement approved and frozen for implementation
IN_REVIEW	Under stakeholder review; implementation may proceed cautiously
DRAFT	Work in progress; not approved for implementation
DEFERRED	Explicitly moved to future release

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1.2 Scope

System Name

NewPOPSys v1.38 - PSP-Led POP Campaign Orchestration Platform

System Overview

NewPOPSys v1 is a modernized campaign orchestration platform that replaces spreadsheet-and-email workflows with an auditable, deterministic lifecycle. The platform enables Print Service Providers (PSPs) to coordinate POP (Point-of-Purchase) campaigns across Brands and Stores with full visibility and control.

Core Platform Loop

Campaign → Store Assignment → PSP Fulfillment → Store Execution → Verification → Visibility

This loop represents the complete lifecycle of a POP campaign from creation through compliance verification.

Included Functionality (v1 Scope)

Module 1: Identity & RBAC

- Multi-tenant architecture with PSP as root tenant
- Brand management within PSP context
- User authentication (email/password)
- Role-based access control with 9 personas
- User invitations and membership management
- API key management for integrations
- Session management and security

Module 2: Stores, Regions & Groups

- Store profile management (1,000+ stores per brand)
- Hierarchical geography (Region > District > Territory)
- Custom store grouping with include/exclude logic
- Store layout and location slot definitions
- External ID aliasing for brand integration

Module 3: Survey Builder & Photo Rules

- Survey template creation with dynamic fields
- Survey versioning with immutable snapshots
- Photo rule configuration per location slot
- Reference image management
- Conditional logic and repeatable sections

Module 4: Campaigns, Kits & Assignment

- Campaign creation (Promotional and Core Branding types)
- Kit definition with item specifications
- Store selection via selection recipes
- Assignment pinning with controlled rebase
- Campaign scheduling and lifecycle management

Module 5: Orders, Shipments & PSP Operations

- Deterministic order generation on campaign publish
- Partial shipment support
- Batch tracking (PRODUCTION, PICK_PACK, SHIP_WAVE, CUSTOM)
- Multi-tracking number support
- PSP MIS integration points

Module 6: Store Execution & Proof Capture

- Mobile-first web application (PWA scaffold)
- Best-effort offline with sync-on-open
- Receive/verify workflow
- Pre-install checklist completion
- Item installation tracking
- Proof photo capture per item/slot
- Completion survey with attestation

Module 7: Verification & Retake Loop

- Photo review queue for Brand/Regional managers
- Approval/rejection with reason codes
- Retake request workflow
- STRICT and FAST verification modes
- Slot-level verification tracking

Module 8: Issues, Reorders & Deinstall

- Issue reporting (Missing, Damaged, Incorrect, Packaging)
- Issue triage and approval workflow
- Reorder generation and tracking
- Campaign expiration handling
- Deinstall task management

Module 9: Exports & Reports

- CSV/XLSX export generation
- PDF report generation
- JSON/XML data exports
- Async export processing with BullMQ

- Export artifact management

Module 10: Integrations & Webhooks

- Signed webhook dispatch
- Inbound API with idempotency keys
- Event outbox pattern for reliability
- Retry scheduling with dead-letter queue
- PSP MIS integration scaffolding

Module 11: Data Retention

- 90-day retention post campaign completion
- Export-before-delete workflow
- Automated purge jobs
- Asset registry for permanent definitions
- Audit event preservation

Module 12: Offline & Sync Strategy

- PWA service worker scaffold
- Offline draft storage
- Sync-on-open submission queue
- Conflict resolution policies

Excluded Functionality (Protected Not-Now)

The following are explicitly **out of scope** for v1 and protected from scope creep:

Not an MIS/ERP

- No job costing or accounting
- No production scheduling
- No payroll integration
- No financial reporting

Not a Marketplace

- No installer bidding/matchmaking
- No vendor marketplace
- No ratings/reviews system
- No dynamic pricing

Not Analytics-Heavy

- No benchmarking dashboards
- No predictive analytics
- No AI insights surfaced to users
- Data readiness only (for future phases)

Not Long-Term Storage

- No archival data warehouse
- No extended retention beyond 90 days
- Export pathways provided for data preservation

Not Endlessly Customizable

- PSP adoption prioritized over store UX flexibility
- Store interface is execution-focused, not configurable

Pilot Context

Dimension	Target
Pilot PSPs	2 (Visual Graphx; Speedy CPS / Alpha Graphics)
Pilot Brands	2-3 (Good2Go confirmed for Speedy/AG)
Stores per Brand	Up to 1,000
Media Volume	>= 1 photo per item per location per campaign
Execution Platform	Web app first; PWA scaffold with offline drafts
Data Retention	90 days post campaign completion

Success Criteria

A campaign can be:

1. Configured, published, and fulfilled without spreadsheets or manual reconciliation
2. Executed by stores with guided workflows and proof capture
3. Verified with approval/rejection/retake cycles
4. Exported with complete audit trail
5. Traced end-to-end using correlationId and audit logs
6. Automatically cleaned up per retention policy

Scope Control

Any scope changes must:

1. Be formally requested through the Decision Log
2. Include tradeoff analysis (what gets deferred)
3. Receive stakeholder approval
4. Update affected SUPP documents
5. Regenerate this SRS with version bump

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1.3 Definitions, Acronyms, and Abbreviations

Abbreviations

Abbreviation	Full Term
API	Application Programming Interface
AWS	Amazon Web Services

BullMQ	Queue library for Node.js based on Redis
CI/CD	Continuous Integration / Continuous Deployment
CSV	Comma-Separated Values
DDL	Data Definition Language
E2E	End-to-End (testing)
ERD	Entity Relationship Diagram
HMAC	Hash-based Message Authentication Code
JSON	JavaScript Object Notation
JSONB	JSON Binary (PostgreSQL data type)
JTBD	Jobs To Be Done (methodology)
LTS	Long-Term Support
MIS	Management Information System
MMS	Mermaid Markdown Syntax
NFR	Non-Functional Requirement
ORM	Object-Relational Mapping
OTLP	OpenTelemetry Protocol
PK	Primary Key
POP	Point of Purchase
PSP	Print Service Provider
PWA	Progressive Web App
RBAC	Role-Based Access Control
REST	Representational State Transfer
S3	Simple Storage Service (AWS)
SDK	Software Development Kit
SKU	Stock Keeping Unit
SLA	Service Level Agreement
SOW	Statement of Work
SQL	Structured Query Language
SRS	Software Requirements Specification
SSO	Single Sign-On

SUPP	Supplemental Specification
TBD	To Be Determined
UI	User Interface
ULID	Universally Unique Lexicographically Sortable Identifier
UUID	Universally Unique Identifier
UX	User Experience
XLSX	Excel Spreadsheet Format
XML	Extensible Markup Language

Domain Terminology

Core Business Entities

Term	Definition
PspTenant	Root tenant representing a PSP organization. The paying customer. All brands, users, and data belong to a tenant.
Brand	A client brand managed by the PSP. Each brand has its own stores, campaigns, and settings.
Store	A physical retail location where POP materials are installed.
Campaign	A brand's promotional program with defined timeline, participating stores, and materials to install.
Kit	The collection of POP items to be shipped and installed for a campaign.
KitItem	A single item within a kit (e.g., "Window Cling 24x36", "Counter Card").

Campaign Types

Term	Definition
Promotional Campaign	Campaign with defined end date that triggers deinstall workflow. Also called "Expiring Campaign".
Core Branding Campaign	Campaign with no end date. Deinstall only triggered manually. Also called "Non-expiring Campaign".

Geographic Hierarchy

Term	Definition
Region	Top level of geographic hierarchy (required). Examples: "West Coast", "Northeast".
District	Second level of geographic hierarchy (optional). Groups stores within a region.
Territory	Third level of geographic hierarchy (optional). Most granular geographic grouping.

StoreGroup	Custom, non-geographic grouping of stores. Used for targeting. A store can belong to multiple groups.
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Survey & Layout

Term	Definition
SurveyTemplate	A reusable survey definition created by Brand Admin. Contains questions, sections, and logic.
SurveyVersion	An immutable snapshot of a SurveyTemplate. Campaigns pin to a specific version.
StoreLayout	Definition of a store's physical advertising locations. Versioned to support changes.
LocationSlot	A specific placement within a store where POP items can be installed.
PhotoRule	Requirements for proof photos at a location slot. Includes min/max photos, resolution, instructions.

Fulfillment

Term	Definition
StoreOrder	An order generated when a campaign is published, representing what PSP needs to produce and ship.
OrderLine	A line item in a StoreOrder specifying SKU and quantity.
Shipment	A physical shipment sent by PSP. An order may have multiple shipments (partial).
ShipmentLine	Items included in a specific shipment. Enables tracking partial fulfillment.
Reorder	A separate order generated to replace missing/damaged items.
Batch	PSP grouping of orders for production efficiency. Types: PRODUCTION, PICK_PACK, SHIP_WAVE, CUSTOM.

Execution & Verification

Term	Definition
StoreAssignment	The record binding a Store to a Campaign. Tracks progress through lifecycle.
ReceiveVerification	Store's confirmation that a shipment was received with item-level acceptance.
CompletionSubmission	Store's submission indicating installation is complete, including attestation.
PhotoUpload	A proof photo uploaded by the store, tied to an item and/or location slot.
PhotoReview	An admin's review decision on a photo (APPROVED/REJECTED with reason).
RetakeRequest	A request for the store to retake a rejected photo.

Issue Management

Term	Definition
IssueRequest	A store-reported problem (missing, damaged, incorrect items).
IssueLine	A specific item and quantity affected by an issue.
IssueType	Categories: MISSING, DAMAGED, INCORRECT, PACKAGING.

Workflow Terms

Term	Definition
Campaign Publish	Action that transitions campaign to active, generates assignments and orders.
Receipt Survey	First-stage store survey upon delivery. Confirms items received, reports issues.
Install Survey	Second-stage store survey. Captures proof photos of installed items.
Completion Attestation	Store user's checkbox confirmation that installation is complete.
Retake Loop	Workflow: rejected photos → RetakeRequest → new upload → review → repeat.
Reorder Flow	Issue approved → Reorder generated → PSP ships → Store receives.
Deinstall Task	End-of-campaign task where store removes expired POP materials.
Waiver	Administrative override marking requirement satisfied without proof.
Reopen	Admin action to unlock a COMPLETE store for additional work.
Split Shipment	Order fulfilled via multiple shipments.

Technical Terms

Term	Definition
Rollup	Aggregation of lower-level quantities to higher levels (Item → Store → Campaign).
Pinning	Freezing a specific version of survey/layout to a store assignment.
Rebase	Updating a pinned version to latest, with re-verification rules.
Idempotency Key	Unique identifier ensuring API request processed only once.
Soft Delete	Marking records as deleted without physical removal.
Event Outbox	Pattern for reliable webhook/event dispatch with retries.

Verification Modes

Term	Definition
STRICT Mode	Every photo requires explicit approval before store completion.
FAST Mode	Photos auto-approve unless flagged. Store completes immediately on submission.

User Roles (Personas)

PSP Level

Role	Definition
Platform Admin	Highest privilege. Full system configuration, tenant management, user impersonation.
PSP Admin	Brand onboarding, PSP-level settings, user management, reporting and exports.
Production Operator	Updates order statuses, creates shipments and tracking, processes batches.

Brand Level

Role	Definition
Brand Admin	Full brand configuration, all campaigns access, store management, user permissions.
Campaign Manager	Brand Admin scoped to assigned campaigns only.
Regional Manager	Exception-first workflow. Reviews photos, handles escalations, scoped to region.

Store Level

Role	Definition
Store Manager	Store-level authority. Manages team, approves replacements, full execution permissions.
Store Operator	Execution-only permissions. Complete surveys, update status, request replacements.

System Level

Role	Definition
Integration User	Service account for API/webhook operations. No UI access.

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1.4 References

Primary Source Documents

Master SOW

Document	Version	Location	Description
MASTER_SOW_COMPILED.md	v1.38	SOW/01_Main_SOW/	Complete compiled v1 specification with all SUPPs as appendices

Shared Foundations SUPPs

Document	Version	Location	Description
SUPP-001	v0.3	SOW/02_SUPPs/Shared_Foundations/	Persona Workflows, JTBD, Screens
SUPP-002	v0.5	SOW/02_SUPPs/Shared_Foundations/	Core Domain Model and State Machines
SUPP-003	v0.7	SOW/02_SUPPs/Shared_Foundations/	RBAC and Permissions Matrix
SUPP-004	v0.5	SOW/02_SUPPs/Shared_Foundations/	Notifications and Escalation Matrix
SUPP-005	v0.3	SOW/02_SUPPs/Shared_Foundations/	Exports, Reports, Output Contracts
SUPP-006	v0.6	SOW/02_SUPPs/Shared_Foundations/	Webhooks and Inbound API Event Contract
SUPP-007	v0.4	SOW/02_SUPPs/Shared_Foundations/	Completion Expiration Overrides
SUPP-008	v0.4	SOW/02_SUPPs/Shared_Foundations/	Data Retention Classification Policy (Option A)

Brand Admin Module SUPPs

Document	Version	Location	Description
SUPP-013	v0.2	SOW/02_SUPPs/Brand_Admin_Module/	Stores, Regions, Groups
SUPP-014	v0.4	SOW/02_SUPPs/Brand_Admin_Module/	Survey Builder, Layout, Photo Rules
SUPP-015	v0.5	SOW/02_SUPPs/Brand_Admin_Module/	Campaigns, Kits, Assignment
SUPP-033	v0.1	SOW/02_SUPPs/Brand_Admin_Module/	Brand Takeout Export Package Scaffold

PSP Operations Module SUPPs

Document	Version	Location	Description
SUPP-016	v0.4	SOW/02_SUPPs/PSP_Operations_Module/	Orders, Shipments, Batches, PSP Ops
SUPP-018	v0.3	SOW/02_SUPPs/PSP_Operations_Module/	Verification, Photo Review, Retake
SUPP-019	v0.3	SOW/02_SUPPs/PSP_Operations_Module/	Issues, Reorders, Expiration, Deinstall

Store Execution Module SUPPs

Document	Version	Location	Description
SUPP-011	v0.3	SOW/02_SUPPs/Store_Execution_Module/	Offline and Sync Strategy
SUPP-017	v0.3	SOW/02_SUPPs/Store_Execution_Module/	Store Execution, Proof Capture

Platform Ops Agent Harness SUPPs

Document	Version	Location	Description
SUPP-012	v0.2	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Technology Selections ADR
SUPP-020	v0.2	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Build Contracts and Physical Schema

SUPP-021	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Repo CI and Vertical Slice Plan
SUPP-022	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Branching, Feature Flags, Release Workers
SUPP-023	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Feature Flags Data Model and Admin UX
SUPP-024	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Env Config, Secrets, Tenant Bootstrap
SUPP-025	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Staging Seed Scale Plan and Testing Targets
SUPP-026	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Local Dev Staging Parity and Toolchain
SUPP-027	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Notifications, Webhooks, Deliverability
SUPP-029	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Observability, Tracing, Logging, Audit
SUPP-030	v0.2	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Admin Ops Console Minimum Screens
SUPP-031	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Supportability, Impersonation, Incident
SUPP-032	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Break-Glass, Retention Safety, Data Requests
SUPP-034	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Rate Limits, Upload Guards, Throttling
SUPP-035	v1.3	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Field Level Data Model, Tables, Fields, Enums
SUPP-038	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Section 508 Accessibility Scaffold
SUPP-039	v0.1	SOW/02_SUPPs/Platform_Ops_Agent_Harness/	Infrastructure, Performance, Security

Screens & Interfaces SUPPs

Document	Version	Location	Description
SUPP-036	v0.6	SOW/02_SUPPs/Screens_Interfaces/	Screens Onboarding and Store Foundation
SUPP-037	v1.0	SOW/02_SUPPs/Screens_Interfaces/	Survey Builder and Store Surveys

Technical Specifications

Database Model

Document	Location	Description
Foundation DDL	SOW/06_Database_Model/01_Schema/	Core schema (tenancy, identity, RBAC)
Module DDLs	SOW/06_Database_Model/01_Schema/	PSP, Brand, Store module schemas
Relationships	SOW/06_Database_Model/02_Relationships/	Foreign key constraints
Indexes	SOW/06_Database_Model/04_Indexes/	Performance optimization indexes

Seed Data	SOW/06_Database_Model/05_Seed_Data/	Enums, system users, default tenant
Validation	SOW/06_Database_Model/07_Validation/	Entity crosswalk and coverage matrix

API Specification

Document	Location	Description
OpenAPI Specification	SOW/06_API/openapi.yaml	RESTful API contract definition
API README	SOW/06_API/README.md	API documentation and usage

Screen Specifications

Series	Location	Description
M-Series (Mobile)	SOW/06_Screen_Specs/	M01-M08: Store mobile app screens
B-Series (Brand)	SOW/06_Screen_Specs/	B01-B07: Brand admin screens
P-Series (PSP)	SOW/06_Screen_Specs/	P01-P03: PSP operations screens
S-Series (Store Portal)	SOW/06_Screen_Specs/	S01-S05: Store portal screens

Design Resources

Wireframes & Prototypes

Resource	URL
Axure Wireframes	https://qvcdv1.axshare.com/
Adobe XD UI Standards	https://xd.adobe.com/view/a639299d-04e9-4f4c-bb70-c89b0d5c8cee-f36a/
Confluence Mind Map	https://servrx.atlassian.net/wiki/spaces/~5f07510d108462001583d6de/pages/14450692/Mind+Map

Local Wireframes

Location	Description
SOW/05_Wireframes/	Interactive HTML prototypes (local copy)
SOW/05_Wireframes/screenshots/	Static screenshot captures
SOW/05_Wireframes/diagrams/	Mermaid state machine diagrams

Reference Documents

Index & Framework

Document	Location	Description
Document Index	SOW/00_Index/00_DOC_INDEX.md	Registry of all documents with versions
SOW Framework	SOW/00_Index/SOW_FRAMEWORK.md	Document development workflow
Glossary	SOW/00_Index/GLOSSARY.md	Domain terminology reference
Process Flow Index	SOW/00_Index/PROCESS_FLOW_INDEX.md	Workflow diagram index
Q&A Session Log	SOW/00_Index/QA_SESSION_LOG.md	Stakeholder decisions

Reference SRS

Document	Location	Description
Ninja v2 SRS	SOW/03_Templates/ninja_docx_extracted/	Reference SRS with IEEE 830 structure

Legacy Reference

Document	Location	Description
PrecisionPOP Legacy	SOW/04_Reference/Legacy_PrecisionPOP/	Legacy system documentation

External Standards

Standard	Description	Relevance
IEEE 830-1998	Software Requirements Specification Standard	Document structure
OpenAPI 3.0	REST API Specification Standard	API contract format
PostgreSQL 15	Database reference documentation	Schema design
WCAG 2.1 AA	Web Content Accessibility Guidelines	Accessibility requirements
Section 508	US Federal accessibility standards	Compliance requirements

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1.5 Document Overview

SRS Structure

This Software Requirements Specification is organized into modular sections following IEEE 830 guidelines, adapted for agentic development workflows. Each section is maintained as separate Markdown files to enable parallel development and version control.

Section Organization

Section 1: Introduction

Establishes the purpose, scope, and context of the SRS document.

File	Content
1.1_Purpose.md	Document objectives, audience, and conventions
1.2_Scope.md	System boundaries, included/excluded functionality
1.3_Definitions_Acronyms.md	Terminology and abbreviations
1.4_References.md	Source documents and external references
1.5_Document_Overview.md	This document structure guide

Section 2: Overall Description

Provides system context and high-level product information.

File	Content
2.1_Product_Perspective.md	System context and external interfaces
2.2_Product_Functions.md	Summary of major system capabilities
2.3_User_Classes.md	Overview of user personas and characteristics
2.4_Operating_Environment.md	Hardware, software, and network requirements
2.5_Constraints.md	Design and implementation constraints
2.6_Assumptions_Dependencies.md	Assumptions and external dependencies
diagrams/system_context.mmd	Mermaid context diagram

Section 3: System Architecture

Documents the technical architecture and data model.

File	Content
3.1_Database_Model.md	Entity-relationship model and schema
3.2_Application_Architecture.md	Component architecture and patterns
3.3_Technology_Stack.md	Technology selections and rationale
3.4_Integration_Architecture.md	External system integrations
diagrams/*.mmd	Mermaid ERD and architecture diagrams

Section 4: User Personas & RBAC

Defines user roles, permissions, and authentication flows.

File	Content
4.1_Persona_Matrix.md	All personas with goals and needs
4.2_Permission_Matrix.md	RBAC grid (persona x feature)
4.3_Authentication_Flows.md	Login, session, security flows
personas/*.md	Detailed persona specifications

Sections 5-10: Module Specifications

Each module section contains screen-by-screen functional requirements.

Section	Module	Screen Series
05_Module_SharedFoundations	Login, Dashboard, Common	L-series
06_Module_MobilePWA	Store Execution App	M-series
07_Module_BrandAdmin	Brand Administration	B-series
08_Module_PSPOperations	PSP Fulfillment	P-series
09_Module_StorePortal	Store Web Portal	S-series
10_Module_PlatformAdmin	System Administration	Admin screens

Each module folder contains:

- X.0_Module_Overview.md - Module summary and navigation
- screens/*.md - Individual screen specifications

Section 11: API Specifications

Documents all API contracts and integrations.

File	Content
11.1_API_Overview.md	API design principles and patterns
11.2_Internal_APIs.md	RESTful endpoint specifications
11.3_External_Integrations.md	Third-party API integrations
11.4_Webhooks.md	Webhook event contracts
schemas/*.json	JSON Schema definitions

Section 12: Non-Functional Requirements

Specifies quality attributes and constraints.

File	Content
12.1_Performance.md	Response times, throughput, capacity

12.2_Security.md	Authentication, authorization, data protection
12.3_Accessibility.md	WCAG 2.1 AA and Section 508 compliance
12.4_Scalability.md	Growth capacity and scaling strategies
12.5_Availability.md	Uptime, redundancy, disaster recovery
12.6_Data_Retention.md	Retention policies and data lifecycle

Section 99: Appendices

Supplementary reference materials.

File	Content
A_State_Machines.md	Campaign, Task, Compliance state diagrams
B_Notification_Matrix.md	Who gets notified when
C_Export_Formats.md	Report and export specifications
D_Glossary.md	Complete terminology reference
E_Change_Log.md	Document revision history

Screen Specification Template

Each screen document follows a consistent structure:

```
# [Screen ID]: [Screen Name]

## 1. Overview
- Screen ID, Module, Primary Persona, SOW Reference

## 2. Wireframe
- Screenshot with link to source

## 3. Screen Description
- Functional description

## 4. User Access (RBAC)
- Persona access matrix with permissions

## 5. Data Model
- Primary entities and relationships (Mermaid ERD)

## 6. APIs
- Internal endpoints and external integrations

## 7. Business Rules
- BR-XXX numbered rules
```

8. State Transitions (if applicable)

- Mermaid state diagram

9. Acceptance Criteria

- AC-XXX numbered criteria

Reading Guide

For Developers

1. Start with **Section 3** (Architecture) to understand the technical foundation
2. Review **Section 4** (Personas/RBAC) for access control context
3. Navigate to the relevant **Module Section (5-10)** for feature implementation
4. Reference **Section 11** (APIs) for endpoint contracts
5. Consult **Section 12** (NFRs) for quality requirements

For AI Development Agents

1. Read **1.1 Purpose** for document conventions and requirement formats
2. Use **1.3 Definitions** for domain vocabulary
3. Reference **1.4 References** to locate source documents
4. Navigate to specific module/screen files for implementation context
5. Cross-reference with SUPP documents for detailed specifications

For QA Engineers

1. Review **1.2 Scope** for included/excluded functionality
2. Use **Module Sections (5-10)** for acceptance criteria per screen
3. Reference **Section 12** (NFRs) for quality attribute testing
4. Consult **Appendix A** (State Machines) for state transition testing

For Product Managers

1. Start with **Section 2** (Overall Description) for product context
2. Review **Section 4** (Personas) for user needs alignment
3. Use **Module Sections (5-10)** for feature scope validation
4. Reference **1.2 Scope** for explicit exclusions

Diagram Conventions

All diagrams use Mermaid syntax for portability:

Diagram Type	Usage
flowchart	Process flows and user journeys
sequenceDiagram	API and system interactions
stateDiagram-v2	Entity lifecycle states
erDiagram	Data model relationships
graph	Architecture components

Traceability

Each requirement traces back to source documentation:

